

Topic 3: HTML Forms, Tables, and Semantic HTML

Learning Objectives

- Build interactive HTML forms with various input types
- Create structured data tables
- Use semantic HTML elements for better page structure
- Understand why semantic HTML matters for accessibility and SEO

3.1 HTML Forms

Forms allow users to enter and submit data. A complete form needs a `<form>` container, input elements, and a submit button.

```
<form action="/submit" method="POST">
  <label for="name">Your Name:</label>
  <input type="text" id="name" name="name" placeholder="Enter name" required>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required>

  <label for="password">Password:</label>
  <input type="password" id="password" name="password" required>

  <button type="submit">Register</button>
</form>
```

Form attributes:

- `action` — URL where form data is sent on submit
- `method` — `GET` (data visible in URL) or `POST` (hidden, used for passwords and private data)

3.2 Common Input Types

```
<!-- Basic text -->
<input type="text" placeholder="Enter text">

<!-- Email - browser validates format -->
<input type="email" placeholder="you@example.com">
```

```

<!-- Password – characters are hidden -->
<input type="password">

<!-- Number with range -->
<input type="number" min="1" max="100">

<!-- Date picker -->
<input type="date">

<!-- Checkbox -->
<input type="checkbox" id="agree" name="agree">
<label for="agree">I agree to the terms</label>

<!-- Radio buttons – only one can be selected -->
<input type="radio" id="male" name="gender" value="male">
<label for="male">Male</label>
<input type="radio" id="female" name="gender" value="female">
<label for="female">Female</label>

<!-- Dropdown -->
<select name="country">
  <option value="pk">Pakistan</option>
  <option value="us">United States</option>
  <option value="uk">United Kingdom</option>
</select>

<!-- Multi-line text -->
<textarea name="message" rows="4" placeholder="Your message here"></textarea>

```

3.3 Labels and Accessibility

Always pair `<label>` with each `<input>` using the `for` attribute:

```

<label for="username">Username:</label>
<input type="text" id="username" name="username">

```

The `for` value must match the input's `id`. This allows:

- Clicking the label to focus the input
- Screen readers to announce what the field is for

3.4 HTML Tables

Tables organize data into rows and columns:

```
<table>
  <thead>
    <tr>
      <th>Student Name</th>
      <th>Grade</th>
      <th>Score</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Ahmed Ali</td>
      <td>A</td>
      <td>92</td>
    </tr>
    <tr>
      <td>Sara Khan</td>
      <td>B+</td>
      <td>87</td>
    </tr>
  </tbody>
</table>
```

Table tags explained:

Tag	Purpose																				
.....																					
<table><tr><td colspan="2">Header section of the table</td></tr><tr><td colspan="2">、</td></tr><tr><td colspan="2">The table container</td></tr><tr><td colspan="2">、</td></tr><tr><td colspan="2">Body section of the table</td></tr><tr><td colspan="2">、</td></tr><tr><td colspan="2">A table row</td></tr><tr><td>、</td><td><table><tr><td>、</td><td>Header cell (bold and centered by default)</td></tr><tr><td>、</td><td>Data cell</td></tr></table></td></tr></table>		Header section of the table		、		The table container		、		Body section of the table		、		A table row		、	<table><tr><td>、</td><td>Header cell (bold and centered by default)</td></tr><tr><td>、</td><td>Data cell</td></tr></table>	、	Header cell (bold and centered by default)	、	Data cell
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、	Data cell																				

Note: Tables should only be used for tabular data, not for page layout.

3.5 Semantic HTML

Semantic elements describe the meaning of content, not just its appearance:

```
<!-- Non-semantic approach (unclear) -->
<div id="header">...</div>
<div id="nav">...</div>
<div id="content">...</div>
<div id="footer">...</div>

<!-- Semantic approach (clear purpose) -->
<header>...</header>
<nav>...</nav>
<main>...</main>
<footer>...</footer>
```

Common semantic elements:

Tag	Purpose
<code><h1></code> <code><h2></code>	Page or section header (logo, site title)
<code><nav></code>	Navigation menu with links
<code><main></code>	Primary content of the page
<code><div></code>	Thematic group of content
<code><article></code>	Self-contained content (blog post, news item)
<code><aside></code>	Side content (sidebar, related links)
<code><footer></code>	Page or section footer
<code></code>	Image or diagram with caption
<code><caption></code>	Caption text for a <code></code>

3.6 Why Semantic HTML Matters

1. **Accessibility** — Screen readers navigate pages using semantic landmarks, helping visually impaired users
2. **SEO** — Search engines understand content structure better, improving search rankings
3. **Readability** — Other developers can read and understand your code faster
4. **Maintainability** — Easier to update styling for specific sections of the page

Key Takeaways

- Forms collect user input and send it to a server
- Use `method="POST"` for sensitive data like passwords
- Always pair `<label>` with `<input>` for accessibility
- Use tables for structured data rows and columns — not for page layout
- Semantic HTML improves accessibility, SEO, and code clarity

Reference Materials

- YouTube: HTML Website Crash Course (freeCodeCamp) — <https://www.youtube.com/watch?v=916GWv2Qs08>
- W3Schools HTML Forms — https://www.w3schools.com/html/html_forms.asp
- W3Schools HTML Tables — https://www.w3schools.com/html/html_tables.asp
- MDN Semantic Elements — https://developer.mozilla.org/en-US/docs/Glossary/Semantics#semantics_in_html

Quiz

Question 1: Which input type makes the browser automatically validate that the user entered a correctly formatted email address?

- A) `type="text"`
- B) `type="mail"`
- C) `type="email"`
- D) `type="address"`

Correct Answer: C — `type="email"` causes the browser to validate the format before allowing form submission. The field must contain an "@" symbol and a domain.

Question 2: Which semantic HTML element should wrap the main navigation links of a website?

- A) `<menu>`
- B) `<navbar>`
- C) `<nav>`
- D) `<header>`

Correct Answer: C — The `<nav>` element is specifically for navigation links. It improves accessibility (screen readers can jump directly to it) and helps search engines understand site structure.
